

# Regional inequality in infant RSV morbidity and mortality burden in the era of expanded RSV passive immunisation: a global projection modelling study

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## Summary

**Background** Respiratory syncytial virus (RSV) is a major cause of acute lower respiratory infection (ALRI) in infants, with low- and middle-income countries (LMICs) bearing a disproportionately high burden. While novel RSV prophylactic products including long-acting monoclonal antibodies and a maternal vaccine are being rolled out for protecting the general infant population in high-income countries, implementation in LMICs is only expected by 2028, potentially exacerbating health inequities. We aimed to assess regional inequality in infant RSV disease burden with and without the implementation of these interventions.

**Methods** We re-analysed the previous global RSV disease burden dataset in infants by WHO regions and World Bank Income Classification for the year of 2019 as a baseline, using generalised linear mixed-effects models. We developed a static deterministic model that explicitly accounted for the varied risks for RSV admission by birth month and chronological age in month and projected the annual RSV-associated ALRI hospital admissions and RSV-attributable deaths under two scenarios: a no-implementation scenario that assumed the same hospital admission and mortality rates to the 2019 baseline, and a June-2025 product use scenario that was based on country-level implementation status of novel RSV prophylactic products as of June 2025; the two scenarios were then compared to estimate the impact of the current implementation of novel prophylactic products. We further calculated the ratio of RSV-attributable deaths between the highest and lowest burden quintiles (Q5-Q1 ratio) to assess inequality under the two scenarios.

**Findings** At baseline, the African Region had the highest incidence of RSV-associated ALRI (117.4 per 1000 person-years, 95% uncertainty range, UR: 60.0–229.8) and highest RSV-attributable mortality rate (0.97 per 1000 person-years, 95% UR: 0.76–1.31) while having the lowest RSV-associated ALRI hospital admission rate (12.6 per 1000 person-years; 95% UR: 7.5–21.3); the Eastern Mediterranean Region had the highest hospital admission rate (29.9 per 1000 person-years, 95% UR: 14.2–63.2). The regional variations in incidence, hospital admission, and mortality rates persisted when further stratifying the above estimates by country income level. As of June 2025, nirsevimab and maternal RSVpreF had been licensed in 54 and 64 countries, respectively, with 16 and 12 countries introducing them into their national immunisation programmes. A total of 4,641,000 doses of nirsevimab and 1,955,000 doses of RSVpreF were estimated to be administered annually under the June-2025 product use scenario globally. Administration of both products was projected to jointly avert 91,000 (95% UR: 75,000–118,000) RSV-associated ALRI hospital admissions and 300 (95% UR: 210–470) RSV-attributable deaths in infants annually, corresponding to 3.7% (95% UR: 2.7–4.9) and 0.4% (95% UR: 0.3–0.7) of the global admissions and deaths, respectively. Approximately 90% of averted hospital admissions and over 70% of averted deaths in infants occurred in high-income countries (81,000 [95% UR: 66,000–108,000] and 220 [95% UR: 140–390], respectively), corresponding to reductions of 33.2% (95% UR 31.0–35.9) and 29.1% (95% UR 22.6–35.2); by contrast, the relative reduction of admissions and deaths in middle-income countries was less than 5%, and no reduction was expected in low-income countries. Globally, substantial inequality was observed in the



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